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Thomas Rigotti, Birgit Schyns and Gisela Mohr
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A Short Version of the Occupational Self-Efficacy Scale: Structural and Construct Validity Across Five Countries

Thomas Rigotti

University of Leipzig

Birgit Schyns

University of Portsmouth

Gisela Mohr

University of Leipzig

Occupational self-efficacy is an important resource for individuals in organizations. To be able to compare the occupational self-efficacy of employees across different countries, equivalent versions of the standard instruments need to be made available in different languages. In this article, the authors report on the structural and construct validity of an instrument that assesses occupational self-efficacy across five countries (Germany, Sweden, Belgium, United Kingdom, Spain), based on an overall sample of $N = 1,535$. The instrument can be recommended for comparative use in German, Swedish, Belgian, Spanish, and British organizational contexts.

Keywords: occupational self-efficacy, validity, cross-national research

Bandura (1977a, 1977b) first introduced the concept of self-efficacy. Self-efficacy is defined as the confidence an individual has in her or his ability to cope with difficult tasks or problems or, as Bandura (1977a) formulated it, is the belief in one's ability to successfully fulfill a task. The interest in self-efficacy has been constant since its introduction in the 1970s. In organizational research, for example, self-efficacy has been shown to directly relate to job satisfaction (Judge & Bono, 2001) and performance (cf. Judge & Bono, 2001; Stajkovic & Luthans, 1998). Self-efficacy can be seen as a personal resource, acting as a buffer in the

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stressor–strain relationship (e.g., Grau, Salanove, & Peiró, 2001; Jerusalem & Schwarzer, 1992). Some authors, however, have also postulated and reported mediating effects of self-efficacy (e.g., Lent & Hackett, 1987) in the stress–strain chain or between personality traits and career interests (Nauta, 2004). This evidence, both theoretical and empirical, highlights the relevance of occupational self-efficacy for work- and organizational-psychology studies and the need for instruments that can be applied in an international context.

As Bandura (1977b) noted, it is important in the assessment of self-efficacy to specify the task that is being addressed. This is based on the idea of specificity matching, meaning that predictor and criterion should have the same level of specificity. Only then it is possible to acquire a good prediction. To illustrate this point, when researchers want to predict a specific behavior, such as writing a scientific paper, they should use a specific self-efficacy—namely, the self-efficacy for writing a scientific paper (cf. Forester, Kahn, & Hesson-McInnis, 2004). Consequently, measures have been developed, in addition to general self-efficacy (cf. Schwarzer, 1994; Shelton, 1990; Sherer et al., 1982), which assess specific domains (e.g., Abele, Stief, & Andra, 2000; Speier & Frese, 1997) or tasks (e.g., Kelly & Nelson, 1999). In organizational research, however, making use of this more specific type of self-efficacy has disadvantages. A comparison between employees in different organizations or between different tasks within one organization can only be done using a broader instrument.

The concept of occupational self-efficacy addresses this point and deals with self-efficacy as a domain-specific assessment. It refers to the competence that a person feels concerning the ability to successfully fulfill the tasks involved in his or her job. The instrument that will be examined in this article has already been used in different language contexts (German: Moser, Schaffner, & Heinle, 2005; Schyns, Paul, Mohr, & Blank, 2005; Chinese: Chin, 2003; and Dutch: Claes et al., 2002). However, we do not yet know the measurement equivalence of different language versions of this instrument. As there is also a need for economic instruments, the aim of this study is to examine the measurement equivalence of a short version of the Occupational Self-Efficacy Scale (Schyns & von Collani, 2002) across five samples from different countries. In addition, we will report validity data on the instrument in these different countries to test whether occupational self-efficacy is embedded within the same correlational pattern. As the countries involved are all from Western Europe, we do not expect possible differences to have a serious impact on the validity of the scale.

Description of the Original Instrument

The instrument introduced by Schyns and von Collani (2002) consists of 20 items derived from different assessments of general self-efficacy as, for example, the General Self-Efficacy Subscale by Sherer et al. (1982). The instrument proved to be related to personality traits, such as general self-efficacy, self-esteem, internal control beliefs, and neuroticism (Schyns & von Collani, 2002) and to organizational outcomes, such as job satisfaction and commitment (Schyns & von Collani, 2002; see

also Schyns et al., 2005). Schyns and von Collani also introduced a short form of the Occupational Self-Efficacy Scale, which is comprised of eight items and shows equally good measurement characteristics as the original long version. In our study, however, we used an even shorter version comprised of six items. The items were selected on the basis of their item characteristics, such as item-total correlation, factor loading, and effect on the internal consistency.

Purpose of the Current Study and Hypotheses

In this study, we will add to prior studies on self-efficacy by comparing the relationships among occupational self-efficacy and performance, job satisfaction and commitment in different countries. In addition, we will examine the relationship between occupational self-efficacy and job insecurity, a variable that has become increasingly important in the recent years of economic downfall in most European countries.

The relationship between self-efficacy and performance has been the topic of many studies. Two meta-analyses showed that self-efficacy is indeed positively related to performance (Sadri & Robertson, 1993; Stajkovic & Luthans, 1998). This is in line with theoretical assumptions that individuals who have a high belief of self-efficacy persist longer (e.g., Multon, Brown, & Lent, 1991) and seek more challenging tasks (e.g., Sexton & Tuckman, 1991), which ultimately leads to higher performance. We, therefore, assume that

Hypothesis 1: Occupational self-efficacy is positively related to performance.

In terms of job satisfaction, we can assume that individuals who feel able and confident concerning their ability to meet the demands of their job will be more satisfied with their job than those who feel that they cannot meet these demands or are barely able to do so. Empirically, Judge and Bono (2001) found a positive relationship between self-efficacy and job satisfaction. More specifically, the Occupational Self-Efficacy Scale that we analyze in this article has been shown to relate to some aspects of job satisfaction (Schyns & von Collani, 2002). Therefore, we assume

Hypothesis 2: Occupational self-efficacy is positively related to job satisfaction.

In line with Schyns and von Collani (2002), we assume that self-efficacious individuals feel attached to the company that provides the job that they feel so able to do. This attachment can also take the form of being unwilling to perform a different task, as it could lead to feelings of insecurity and a fear of not being able to cope. Contrarily, individuals who feel incompetent on the job are not likely to emotionally attach to their tasks and their organization. This would suggest that self-efficacy and affective commitment are related. The relationship between self-efficacy and affective commitment has, in fact, been empirically supported on a general level (Bozeman, Perrewe, Hochwarter, & Brymer, 2001; Tracey, Hinkin, Tannenbaum, & Mathieu, 2001) and on a more specific level,

in the case of the Occupational Self-Efficacy Scale (Schyns et al., 2005; Schyns & von Collani, 2002). Therefore, our third hypothesis reads:

Hypothesis 3: Occupational self-efficacy is positively related to commitment.

It can be assumed that individuals who feel less competent than others concerning their abilities with respect to their job will experience a sense of job insecurity (cf. Söhnlein, 1998). Therefore, we assume that

Hypothesis 4: Occupational self-efficacy is negatively related to job insecurity.

Based on these assumptions and prior results, self-efficacy can be seen as an important resource for individuals in organizations, making further study of the concept sensible. The growing interest in cross-cultural studies, as well as the cultural diversity of employees in “global player” enterprises, call for reliable, valid, and comparable instruments in different languages. For this reason, we will compare the results of our hypotheses across different countries. Given that the countries involved (Germany, Sweden, Belgium, United Kingdom, and Spain) have cultural similarities and, as members of the European Union, economic similarities, we do not expect significant differences in the relationship between self-efficacy and outcomes in these countries. To put it even stronger, similarities in the relationships across countries would support the validity of the instrument on which we are focusing.

As individuals may differ in their self-efficacy along sociodemographic characteristics (e.g., for gender differences in general self-efficacy, see Scholz, Gutiérrez Doña, Sud, & Schwarzer, 2002; for gender differences in task-specific occupational self-efficacy, see Koumoundourou, 2004; and for differences related to occupational experience, see Abele et al., 2000), we will control for possible effects of sample composition by testing for effects of gender, age, education, and weekly working hours on occupational self-efficacy.

METHOD

Samples

The samples are part of a pilot study of the European research project PSYCONES. Questionnaires were distributed predominantly in three sectors: health care (mainly hospitals), retail shops, and temporary work agencies. In the Belgian and British samples, employees from manufacturing plants also participated in the study.

As our primary goal was to evaluate the Occupational Self-Efficacy Scale, we performed a listwise deletion on the missing values of these items. The highest rate of missing values was observed in the Belgian sample, with 3.5%. However,

such small proportions of missing values are not expected to bias the analyses in a serious manner (Tabachnick & Fidell, 2001, p. 59).

Overall, our sample consists of persons from five countries ($N = 1,535$). Women are overrepresented, except in the British sample. Country samples differ in their composition and show different characteristics in terms of the percentage of sampled individuals from different sectors, $\chi^2(24, N = 1,535) = 1921.84, p < .001$; the share of males and females, $\chi^2(4, N = 1,535) = 91.59, p < .001$; the age of participants, $F(4) = 22.00, p < .001$; and the educational level, $\chi^2(16, N = 1,535) = 505.52, p < .001$. The average number of working hours a week also differed, $F(4) = 8221.64, p < .001$; see Table 1).

Instruments

Prior to distribution, all scales were translated into the target language by independent bilingual teams, using the original language version as reference and using a translation-back translation technique (cf. Behling & Law, 2000; Werner & Campbell, 1970). Furthermore, we performed factor analyses for all employed scales in all samples to support factorial validity and to be sure that items are reflecting one latent construct.

Sociodemographics. Age was assessed as a continuous variable. Gender is a dichotomous variable with 0 for females and 1 for males. Educational level was measured by using a country-specific combination of school and vocational education, which was then positioned on the six levels of the international standard scheme International Standard Classification of Education (ISCED; see Organisation for Economic Cooperation and Development, 1999). Weekly working hours were assessed as an open question, resulting in a continuous variable.

Occupational self-efficacy. The long version of the Occupational Self-Efficacy Scale developed by Schyns and von Collani (2002) consisted of 20 items taken from four different scales: the General Self-Efficacy Scale developed by Sherer et al. (1982), the Generalized Self-Efficacy Scale developed by Schwarzer (1994), the Hope Scale developed by Snyder et al. (1991), and the Heuristic Competence Scale developed by Stäudel (1988). The items were then adapted and reformulated for the work context. A shortened version consisting of eight items was found to be a reliable and valid measure in a German sample (Schyns & von Collani, 2002). In the present study, six of these eight items were selected on the basis of prior scale and item analyses of German applications. Items could be rated on a six-level response scale ranging from 1 (*not at all true*) to 6 (*completely true*). High values reflect high occupational self-efficacy.

Job satisfaction. Four items were taken from Price (1997) to measure job satisfaction (e.g., "I find enjoyment in my job") and could be assessed on a range from 1 (*completely disagree*) to 5 (*completely agree*). The internal consistencies in the different language versions range from .77 to .90.

Table 1
Sample Characteristics

	N	Health Care (%)	Retail (%)	Temporary Agency Work (%)	Industry (%)	Males (%)	Age		Education		Weekly Working Hours	
							M	SD	M	SD	M	SD
Germany	200	34.8	31.9	33.4	—	28.1	34.52	11.15	2.42	1.00	36.73	8.87
Sweden	147	—	29.8	70.2	—	12.9	40.93	10.83	4.09	1.42	33.31	9.58
Belgium	616	49.5	6.1	5.9	38.5	38.1	36.20	10.26	3.53	0.98	36.35	7.12
Great Britain	195	—	29.9	—	65.0	56.1	35.07	11.51	3.88	0.94	33.83	11.59
Spain	377	51.4	25.7	22.9	—	25.7	32.29	6.77	3.60	5.11	40.33	9.01
Total	1,535	37.1	19.5	18.8	24.6	33.4	35.35	10.17	3.50	2.71	36.77	9.01

Note: Educational level was coded according to the International Classification of Education, 1 = primary education, 2 = lower secondary education, 3 = upper secondary education, 4 = post-secondary, non-tertiary education, 5 = first stage of tertiary education, and 6 = second stage of tertiary education.

Commitment. The Organizational Commitment Scale by Cook and Wall (1980) assesses the affective relationship of the employee to his employing organization. Commitment is defined as the “feelings of attachment to goals and values of the organisation, one’s role in relation to this, and attachment to the organisation for its own sake rather than for its strictly instrumental value” (Cook & Wall, 1980, p. 40). The scale consists of four items (e.g., “I am quite proud to be able to tell people who it is I work for”) with a 7-point, Likert-type scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The internal consistencies in the various language versions range from .72 to .85.

Performance. A nine-item measure from Abramis (1994) was used to measure perceived performance. We asked participants to give a self-evaluation of performance by thinking about the last working week (e.g., “In your own judgment, how well did you fulfill the following tasks? Finish things on time”). Responses ranged from 1 (*very badly*) to 5 (*very well*). The internal consistencies in the various language versions range from .79 to .88.

Job insecurity. A four-item measure developed by De Witte (2000) was used to assess job insecurity (e.g., “I feel insecure about the future of my job”). The response format ranged from 1 (*strongly disagree*) to 5 (*strongly agree*). The internal consistencies in the different language versions range from .77 to .86.

All measures have been used in numerous studies and were also used in a later multinational study (Rigotti, Mohr, Clinton, & Guest, in press). Hence, there is broad support for their construct validity.

Analyses to Test for Measurement Equivalence Across Groups

Two different methodological paradigms are generally suggested for the test of measurement equivalence across different groups or situations: item response theory (IRT), and confirmatory factor analyses (CFA; cf. Meade & Lautenschlager, 2004; Raju, Lafitte, & Byrne, 2002; Schriesheim & Neider, 2001). Recent discussions propose the application of both methods, as they both have their pros and cons (e.g., Mead & Lautenschlager, 2004). However, IRT needs big sample sizes even more than CFA does. IRT also demands a certain number of items to come to valid results. CFA has the advantage of being able to compare multiple groups in one analysis (cf. Mead & Lautenschlager, 2004). These considerations led us to the decision to test the measurement equivalence of the Occupational Self-Efficacy Scale with single- and multiple-group CFA within the framework of SEM (structural equation modeling). CFA assumes indicators or manifest variables (i.e., items) to be linearly related to one or more unobserved latent constructs (cf. Jöreskog & Sörbom, 1984). This measurement model includes the estimation of factor loadings, variance-covariance between latent factors, and measurement errors. Programs to conduct multiple-group analyses (we used Amos 5.0; Arbuckle, 2003) allow for the fixing of certain parameters so that restrictions can be imposed on the model. In

general, the proposed sequence of restrictions is: invariance of factor loadings, equality of variance–covariance between latent factors, and equal measurement errors (e.g., Van de Vijver & Leung, 1997).

Scholars differ in their criteria for measurement invariance. Some have proposed invariant factor loadings as sufficient criteria (e.g., Drasgow & Kanfer, 1985; Reise, Widaman, & Pugh, 1993; Salzberger, Sinkovics, & Schlegelmilch, 1999). “An item or a subscale is said to display measurement equivalence . . . if the item factor loadings are invariant across two populations” (Raju et al., 2002, p. 519). Others have argued the need for invariant measurement errors (e.g., Byrne & Watkins, 2003), and some have even claimed equal intercepts between groups or situations to be necessary (e.g., Chan, 2000).

We will test for measurement equivalence by using multiple-group comparisons with nested models. Out of the variety of possible fit indices, we will present the χ^2 statistics, the goodness-of-fit Index (GFI; Jöreskog & Sörbom, 1984), the adjusted goodness-of-fit index (AGFI; Jöreskog & Sörbom, 1984), the comparative fit index (CFI; Bentler, 1990) and the 90% confidence interval (CI) of the root mean square error approximation (RMSEA; Steiger, 1989). Although values greater than .90 are considered to represent a good fit in terms of the GFIs, and values greater than .80 are considered acceptable, the RMSEA should be less than .05 for a good fit and less than .08 for a still reasonable fit of the data to the model (Browne & Cudeck, 1993). As $\Delta\chi^2$ has its critics because of its dependency on sample size (e.g., Cheung & Rensvold, 2002; Hu & Bentler, 1998; Kelloway, 1995), we also used Δ CFI as an indicator in the comparison of models. According to Cheung and Rensvold (2002), values greater than .01 indicate a significant drop in fit (cf. Lievens & Anseel, 2004).

RESULTS

Single-Group Factorial Validity

CFA (Amos 5.0; Arbuckle, 2003) with maximum-likelihood estimation was used to test factorial validity. Item 3 was used as a reference indicator. The loading of this item was set to 1.0 (cf. Bollen, 1989) because this item showed the highest factor loadings in all versions. Table 2 presents the item communalities, factor loadings (maximum-likelihood), item–total correlations, and FIT statistics for the five country samples. The GFI and CFI show values greater than .90 for all versions. The AGFI, with values between .83 and .86 (.94 in the Spanish sample), still indicates an acceptable model fit. Only the RMSEA points toward some redundancy in the item pool, with the exception of the Swedish and Spanish versions (see Table 2). If item 5 is omitted to meliorate the fit, AGFI values are then greater than .90 in all but the British sample, and RMSEA values are less than the .08 criteria (also except for the British sample). However, internal consistencies would then slightly drop by

Table 2
Communalities, Factor Loadings, Item–Total Correlations, Cronbach’s Alpha and Fit Statistics for
Confirmatory Factor Analyses of Occupational Self-Efficacy Across Five Country Samples

Items	Germany (N = 200)			Sweden (N = 147)			Belgium (N = 616)			Britain (N = 195)			Spain (N = 377)		
	h^2	f	r_{it}	h^2	f	r_{it}	h^2	f	r_{it}	h^2	f	r_{it}	h^2	f	r_{it}
1. I can remain calm when facing difficulties in my job because I can rely on my abilities.	.56	.75	.66	.54	.74	.66	.58	.76	.65	.60	.78	.72	.56	.75	.68
2. When I am confronted with a problem in my job, I can usually find several solutions.	.66	.81	.73	.58	.76	.68	.55	.74	.65	.59	.77	.74	.54	.74	.67
3. Whatever comes my way in my job, I can usually handle it.	.67	.82	.74	.55	.74	.67	.70	.84	.73	.69	.83	.77	.59	.77	.70
4. My past experiences in my job have prepared me well for my occupational future.	.52	.72	.68	.45	.67	.62	.30	.55	.53	.46	.68	.65	.46	.68	.63
5. I meet the goals that I set for myself in my job.	.33	.57	.56	.41	.64	.60	.34	.58	.58	.60	.78	.73	.45	.67	.62
6. I feel prepared for most of the demands in my job.	.42	.65	.63	.53	.73	.68	.48	.70	.66	.63	.79	.73	.49	.70	.65
Cronbach’s alpha			.87			.86			.85			.90			.86
$\chi^2/df = 9$	4.77		2.94			14.05			5.36			3.43			
CFI	.93		.94			.93			.93			.97			
AGFI	.84		.86			.86			.83			.94			
CFI	.94		.95			.92			.94			.98			
90% CI RMSEA	.10 to .18		.07 to .17			.12 to .17			.11 to .19			.05 to .11			
RMSEA p values	.00		.02			.00			.00			.05			

Note. Estimation method: maximum likelihood. h^2 = communalities; f = factor loadings; r_{it} = item–total correlations; GFI = goodness of fit; AGFI = adjusted goodness of fit; CFI = comparative fit index; RMSEA = root mean square error of approximation; CI = confidence interval.

.01 to .03 points. We therefore decided to hold on to the six-item version of the Occupational Self-Efficacy Scale.

Reliability coefficients for the Occupational Self-Efficacy Scale are between .85 (Belgium) and .90 (Great Britain) and support a good internal consistency of the scale. The item–total correlations are greater than .50 for all items and samples.

Equivalence of Language Versions

When testing the unconstrained model of the Occupational Self-Efficacy Scale across all five country samples, the multiple-group comparison showed a good fit, indicating that a one-dimensional model can be accepted for all language versions (see Table 3). Under the constraint of equal factor loadings (measurement weights), a significant increase of the chi-square statistic could be observed, although all other fit statistics point toward a satisfying fit for the multiple-group comparison. The assumption of equal factor variances is also supported by the ΔCFI . The invariance of error-variance across country samples is, however, not supported in the multiple-group comparison ($\Delta\text{CFI} > .01$). In sum, these findings support configural invariance (basic factor patterns correspond) as well as metric invariance (factor loadings correspond). Constraints have to be made only concerning scalar invariance (relationship of latent and manifest variables correspond), which is problematic for testing simple mean differences between country samples. To achieve more detailed information about the equivalence of the language applications in comparison to the original German version, we also performed two-group comparisons. The results of these analyses are also summarized in Table 3. It can be observed that equal factor loadings in comparison to the German sample can be found in all other samples, indicated by ΔCFI values lower than .01. The assumption of equal factor variances is not supported only for the comparison of the German with the Swedish sample ($\Delta\text{CFI} = .019$). Invariance of residuals is supported in the German–Spanish comparison ($\Delta\text{CFI} = .009$), and seems still to be acceptable for the German–Belgian comparison ($\Delta\text{CFI} = .016$) but finds no support in the Swedish–German and British–German comparisons.

Construct Validity

We will first present the results concerning the relationship between occupational self-efficacy and sociodemographic variables and then present the results of the partial correlation analyses (controlling for the sociodemographic variables that were significantly related to occupational self-efficacy) separately for all language versions, using the occupational self-efficacy measure and measures of job satisfaction, commitment, performance, and job insecurity (see Table 4).

Sociodemographics and occupational self-efficacy. No gender differences could be observed within country samples (t tests for independent samples). Age, which can be seen as an indicator of occupational experience, could be shown to be positively related to occupational self-efficacy in the German ($r = .28, p < .001$), Belgian

Table 3
Multiple-Group and Two-Group Comparisons of Measurement Invariance

		<i>df</i>	χ^2	Δdf	$\Delta\chi^2$	GFI	AGFI	CFI	ΔCFI	RMSEA	90% CI	<i>p</i> ^a
All groups	Unconstrained	45	274.90	—	—	.94	.86	.94	—	.05	.06	.02
	Equal factor loadings	65	315.62	20	40.72**	.93	.89	.94	.006	.05	.06	.47
	Equal factor variance	69	337.45	4	21.83**	.93	.89	.93	.004	.05	.06	.44
	Equal error variance	93	518.74	24	181.29**	.90	.88	.89	.040	.05	.06	.04
Germany versus Sweden	Unconstrained	18	69.34	—	—	.94	.85	.94	—	.07	.11	.00
	Equal factor loadings	23	78.07	5	8.73	.93	.87	.94	.004	.06	.10	.00
	Equal factor variance	24	95.52	1	17.45**	.91	.85	.92	.019	.07	.11	.00
	Equal error variance	30	197.94	6	102.42**	.85	.79	.82	.097	.11	.14	.00
Germany versus Belgium	Unconstrained	18	169.35	—	—	.93	.84	.93	—	.09	.12	.00
	Equal factor loadings	23	187.52	5	18.17**	.92	.86	.92	.005	.08	.11	.00
	Equal factor variance	24	195.25	1	7.73**	.92	.86	.92	.003	.08	.11	.00
	Equal error variance	30	234.60	6	39.35**	.90	.86	.90	.016	.08	.10	.00
Germany versus Spain	Unconstrained	18	73.78	—	—	.96	.90	.96	—	.06	.09	.01
	Equal factor loadings	23	81.34	5	7.56	.95	.91	.96	.002	.05	.08	.04
	Equal factor variance	24	83.32	1	1.98	.95	.92	.96	.000	.05	.08	.05
	Equal error variance	30	101.91	6	18.59**	.94	.91	.95	.009	.05	.08	.04
Germany versus Britain	Unconstrained	18	91.12	—	—	.93	.83	.94	—	.08	.12	.00
	Equal factor loadings	23	112.14	5	21.02**	.91	.84	.93	.013	.08	.12	.00
	Equal factor variance	24	112.93	1	0.79	.91	.84	.93	.000	.08	.12	.00
	Equal error variance	30	161.89	6	48.96**	.88	.83	.84	.036	.09	.12	.00

Note. GFI = goodness of fit; AGFI = adjusted goodness of fit; ΔCFI = comparative fit index; RMSEA = root mean square error of approximation; CI = confidence interval.

a. This is the *p* value for the RMSEA, also referred to as the “pclose.”

p* < .05. *p* < .01.

Table 4
Partial Correlations Between Occupational Self-Efficacy and a Set of Variables (Controlling for Age)

Occupational Self-Efficacy	N	Partial Correlation Controlling for Age											
		Job Satisfaction			Commitment			Performance			Job Insecurity		
		r	Fz	α	r	Fz	α	r	Fz	α	r	Fz	α
Germany	191 to 200	.17*	—	.84	.19**	—	.85	.32**	—	.79	-.19**	—	.77
Sweden	136 to 144	.46**	2.95**	.84	.39**	1.99*	.72	.58**	3.00**	.79	-.27**	0.77	.77
Belgium	579 to 611	.32**	1.95	.88	.20**	0.13	.80	.43**	1.56	.79	-.09*	1.25	.82
UK	192 to 195	.19**	0.20	.90	.13	0.61	.76	.44**	1.39	.88	.06	2.49**	.86
Spain	367 to 374	.23**	0.71	.77	.16**	0.35	.82	.45**	1.74	.82	-.11*	0.93	.79

Note. r = partial correlation (controlling for age); Fz = Fisher's z test comparing Germany to other countries (two-tailed test).
* $p < .05$. ** $p < .01$.

($r = .10, p < .01$), and British ($r = .18, p < .05$) samples. Mean differences across educational levels only appeared to be significant within the Swedish sample (ANOVA: $F(4) = 5.11, p < .01$), and a post hoc Scheffé test could show that this is only because of differences in the two most distinct groups (primary education versus college degree). Finally, weekly working hours did not prove to be related to occupational self-efficacy in any country sample. When comparing country samples in the following section, we will thus only control for age as a potential confounding variable.

Occupational self-efficacy in relation to other constructs. In accordance with the findings reported in previous studies, we expected occupational self-efficacy to be positively related to the two job-related attitudes: job satisfaction and commitment. Substantial partial correlations (controlled for age) could be found in all samples for job satisfaction. Significant partial correlations were also found for commitment across all country samples, with the exception of Great Britain, although effect sizes are rather low (see Table 4).

Strong evidence for the relatedness of occupational self-efficacy and perceived performance could be found, with significant partial correlations displaying medium to large effect sizes across all samples. In addition, we tested the assumption that occupational self-efficacy is negatively related to perceived job insecurity. In all samples except the British one, we found evidence for this assumption, although effect sizes are small.

Furthermore, we performed bivariate correlation comparisons between the German sample and the four other country samples using Fisher's z test. The z values are also displayed in Table 4. The critical values are $z > 1.96$ for $p < .05$, and $z > 2.58$ for $p < .01$ (two-tailed). Correlation coefficients are significantly higher in the Swedish sample than in the German sample for the relationship between occupational self-efficacy and job satisfaction, commitment, and performance, whereas the correlation coefficients from the Belgian, British, and Spanish samples do not significantly differ from the German results. There is one significant correlation difference with respect to job insecurity, in which a significantly lower coefficient can be found in the British sample as compared to the German sample.

DISCUSSION

Overall, the application of the Occupational Self-Efficacy Scale can generally be recommended for the analyzed German, Swedish, Belgian, British, and Spanish applications. When we acknowledge the interdependence of the χ^2 statistic with the sample size, all tested versions can be considered to be metrically equivalent to the German version. As invariance of error variances could not be supported in the comparison of the German with the Swedish and British samples, one might want to be cautious about making mean comparisons between

applications of these versions. For correlational studies, however, the evidence of psychometric invariance seems to be sufficient for all presented language versions.

First, evidence for the construct validity could be reported. Results were in line with previous empirical studies. The stronger relationship between occupational self-efficacy and job-related attitudes in the Swedish sample may be related to sample characteristics (e.g., the large share of temporary agency workers).

Limitations

For the purpose of this study, we used convenient samples from a European study. Samples differed along sociodemographic characteristics, which may hamper the comparability of different language versions. However, participants in our study were all working employees and not students, as in most other studies dealing with occupational or general self-efficacy (e.g., Abele et al., 2000; Koumoundouros, 2004; Schwarzer, Bäßler, Kwiatek, & Schröder, 1997, among others). We would like to stress that this fact may also alter the external validity of results. As our results are based on cross-sectional data, we cannot draw any causal conclusions regarding the role of occupational self-efficacy as a process of occupational socialization. Furthermore, common method variance may be seen as a source of bias in questionnaire-based correlational studies (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). However, neither of these issues was the focus of our study, which was the comparison of the application of the Occupational Self-Efficacy Scale across countries to examine the cross-cultural reliability and validity of the instrument.

Conclusions and Outlook

The questions concerning the interrelation between general, task-specific, or domain-specific measures of self-efficacy, as well as the question concerning whether the effects of these measures can function as predictors of behavior in organizations or in general, are intriguing and are not yet fully answered (e.g., Mudgett & Quiñones, 1997; Woodruff & Cashman, 1993). However, compared to task-specific measures, occupational self-efficacy as a domain-specific measure is seen as a rather broad construct, "so that employees in different organizational contexts can be compared" (Schyns & von Collani, 2002, p. 237). Moreover, compared to generalized self-efficacy, occupational self-efficacy has the advantage of being more applicable in an organizational context.

We recommend the use of the Occupational Self-Efficacy Scale for occupational health studies (as an important internal resource that may buffer strain effects of stressors), for studies dealing with employment transitions (e.g., from vocational training, unemployment or temporary employment to permanent employment), and in vocational contexts as a possible evaluation criterion of training programs. We hope that the assessment of occupational self-efficacy described here will motivate further investigations in this field, especially studies in a cross-national setting.

APP ENDIX

Items of the Short Occupational Self-Efficacy Scale in Five Languages

Item	German	Swedish	Belgian	British	Spanish
1	Beruflichen Schwierigkeiten sehe ich gelassen entgegen, weil ich mich immer auf meine Fähigkeiten verlassen kann.	Jag kan behålla lugnet när jag möter svårigheter i mitt arbete eftersom jag litar på min egen förmåga.	Ik kan kalm blijven wanneer ik geconfronteerd word met moeilijkheden in mijn job, omdat ik kan terugvallen op mijn vaardigheden.	I can remain calm when facing difficulties in my job because I can rely on my abilities.	Permanezco tranquilo cuando me enfrento a dificultades en mi trabajo porque confío en mis capacidades.
2	Wenn ich bei der Arbeit mit einem Problem konfrontiert werde, habe ich meist mehrere Ideen, wie ich damit fertig werde.	När jag möter problem i arbetet kan jag oftast hitta flera olika lösningar.	Wanneer ik geconfronteerd word met een probleem in mijn werk, dan vind ik meestal verschillende oplossingen.	When I am confronted with a problem in my job, I can usually find several solutions.	Cuando afronto un problema en mi trabajo, normalmente encuentro varias soluciones.
3	Was auch immer in meinem Berufsleben passiert, ich werde schon klarkommen.	Vad som än händer i arbetet så har jag oftast förmåga att hantera det.	Wat er ook gebeurt in mijn werk, ik kan het gewoonlijk wel aan.	Whatever comes my way in my job, I can usually handle it.	Normalmente puedo manejar cualquier problema que se cruce en mi camino o en mi trabajo.
4	Durch meine vergangenen beruflichen Erfahrungen bin ich gut auf meine berufliche Zukunft vorbereitet.	Mina tidigare arbetserfarenheter har gjort mig väl förberedd för mitt kommande yrkesliv.	De ervaringen die ik in het verleden in mijn job heb opgedaan, hebben me goed voorbereid op mijn beroep in de toekomst.	My past experiences in my job have prepared me well for my occupational future.	Mis experiencias pasadas en mi trabajo me han preparado bien para mi futuro laboral.
5	Ich erreiche die beruflichen Ziele, die ich mir setze.	Jag brukar klara av de mål jag sätter upp för mig själv i arbetet.	Ik haal de doelstellingen die ik aan mezelf stel in mijn job.	I meet the goals that I set for myself in my job.	Logro las metas que me propongo a mi mismo en mi trabajo.
6	Ich fühle mich den meisten beruflichen Anforderungen gewachsen.	Jag är väl förberedd för de flesta krav jag kan möta i mitt arbete.	Ik ben voldoende gewapend om de eisen van mijn job het hoofd te bieden.	I feel prepared for most of the demands in my job.	Me siento preparado para la mayoría de las demandas de mi trabajo.

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